

SolarGen Production Forecast Model

SolarGen

Model version: production OM-weighted blend selected on paired history through
2026-06-12

1 Scope

SolarGen forecasts rooftop photovoltaic generation for a 10 kWp south-facing rooftop array in OHZ / Osterholz-Scharmbeck. The production forecast shown in the browser and in the local history app is an Open-Meteo-weighted blend of an Open-Meteo current physical forecast and a DWD stable forecast.

The browser app obtains DWD data through the Open-Meteo DWD ICON endpoint. The local history app obtains DWD data from DWD MOSMIX. Both surfaces use the same transfer structure and final production blend.

2 Source Physical Model

For each source $s \in \{\text{OM}, \text{DWD}\}$, hourly meteorological inputs are:

$$I_{s,h}, C_{s,h}, R_{s,h}, T_{s,h},$$

where I is global tilted irradiance in W/m^2 , C is cloud cover, R is precipitation, and T is ambient temperature.

The source physical model computes a rooftop generation curve:

$$G_{s,h}^{\text{cur}} = f_{\text{site}}(I_{s,h}, C_{s,h}, R_{s,h}, T_{s,h}).$$

The daily current total for source s is:

$$G_s^{\text{cur}} = \sum_{h=0}^{23} G_{s,h}^{\text{cur}}.$$

3 DWD Stable Transfer

The DWD current physical model alone did not generalize well. SolarGen therefore transforms DWD through a stable daily transfer model before production blending.

First compute a DWD sunshine/rain daily model:

$$G_{\text{DWD}}^{\text{raw}} = \max(0, 18.3545 + 2.3510H_{\text{DWD}} - 1.9219R_{\text{DWD}}^+),$$

where H_{DWD} is DWD sunshine duration in hours and R_{DWD}^+ is DWD precipitation summed only over hours with positive tilted irradiance:

$$R_{\text{DWD}}^+ = \sum_{h=0}^{23} \mathbf{1}\{I_{\text{DWD},h} > 0\} R_{\text{DWD},h}.$$

The stable DWD daily forecast is:

$$G_{\text{DWD}}^{\text{stable}} = 0.25G_{\text{DWD}}^{\text{cur}} + 0.75G_{\text{DWD}}^{\text{raw}} + 4.039$$

4 Production Blend

The production daily forecast is:

$$G^{\text{prod}} = 0.73G_{\text{OM}}^{\text{cur}} + 0.27G_{\text{DWD}}^{\text{stable}}$$

There is no production-bias term in the current model.

5 Hourly Allocation

The DWD current hourly curve is scaled to the stable DWD daily total:

$$\tilde{G}_{\text{DWD},h} = G_{\text{DWD},h}^{\text{cur}} \cdot \frac{G_{\text{DWD}}^{\text{stable}}}{\sum_{j=0}^{23} G_{\text{DWD},j}^{\text{cur}}}.$$

The production hourly curve is:

$$G_h^{\text{prod}} = 0.73G_{\text{OM},h}^{\text{cur}} + 0.27\tilde{G}_{\text{DWD},h}.$$

The curve is then scaled by a small numerical factor if needed so that:

$$\sum_{h=0}^{23} G_h^{\text{prod}} = G^{\text{prod}}.$$

Any rounding adjustment is applied to the largest positive production hour so that night hours cannot become negative.

6 Curtailment, Load, Battery, And Value

The browser app applies the household load, battery, export, and value model to G_h^{prod} . For each hour:

$$G_h^{\text{curt}} = \max(0, G_h^{\text{prod}} - F),$$

$$G_h^{\text{deliv}} = G_h^{\text{prod}} - G_h^{\text{curt}},$$

where $F = 6$ kW by default. Delivered PV first covers same-hour household load, then charges the battery, and remaining surplus is exported.

7 Selection Evidence

The production model was selected on paired Open-Meteo and DWD day-ahead forecast history with actuals through 2026-06-12.

Model	In-sample MAE	LOO MAE	Rolling MAE	Comment
Equal blend	3.918	fixed	5.010	recent low bias
OM only	4.147	fixed	–	single source
DWD stable only	4.567	fixed	–	single source
Fitted weight, no bias	3.861	3.945	5.208	selected
Fitted weight plus bias	3.844	4.485	5.419	overfits

The selected model is not the best in-sample model. It is the smallest recalibration that improves the retained sample while avoiding a bias term that degraded out-of-sample checks.

8 Current Stored-History Accuracy

After recomputing production rows under the recalibrated blend, the local history database contains 30 paired actual days:

Metric	Value
MAE	3.861 kWh
RMSE	5.186 kWh
Bias	1.185 kWh
MAPE	10.69 percent

9 Implementation References

- `src/model.js`: source physical model.
- `src/productionBlend.js`: browser production blend.
- `src/weather.js`: Open-Meteo and DWD ICON browser endpoints.
- `history_app/forecast_model.py`: local production blend using stored OM/DWD inputs.
- `history_app/cli.py`: `recompute-production` command.

10 Maintenance Rule

Whenever model parameters change, update this document, regenerate the PDF, recompute production history with:

```
python3 -m history_app.cli recompute-production
```

and run:

```
npm run check    npm test
```